$x_1, x_2, \ldots, x_n$ i.i.d. $x_i \sim \left(\frac{3}{2} + 4\theta\right) x^{4\theta + \frac{1}{2}}$

EXERCISE 4

a) For $P(x;\theta)$ to be a pdf we impose the condition

$$\int_0^1 (k + 4\theta)\, x^{4\theta + \frac{1}{2}}\, dx = 1$$

$$= (k+4\theta)\left[\frac{x^{4\theta + \frac{1}{2} + 1}}{4\theta + \frac{1}{2} + 1}\right]_0^1 = \frac{k + 4\theta}{4\theta + \frac{1}{2} + 1} = 1$$

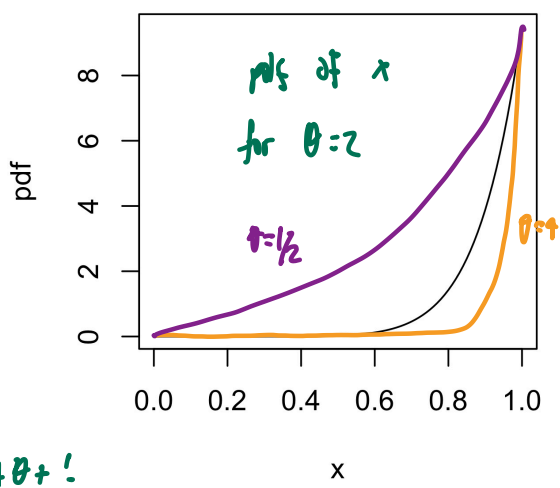
$$\Leftrightarrow \quad k + \cancel{4\theta} = \cancel{4\theta} + \frac{1}{2} + 1 \quad \Leftrightarrow \quad k = \frac{3}{2}$$

b) Use R-Studio

```
theta = 2
k = 3/2
x = seq(0,1,0.01)
pdf = (k+4*theta)*x^(4*theta+1/2)
plot(x,pdf,type="l")
```

try different values

sketch by hand



pdf evaluated at x_1

$$P(x_1;\theta) = \left(\frac{3}{2} + 4\theta\right) x_1^{4\theta + 1/2}$$

$$P(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} \left(\frac{3}{2} + 4\theta\right) x_i^{4\theta + \frac{1}{2}}$$

3) (MAIN POINT)

Given $x_1, \dots, x_n$ i.i.d. we have

$$\ell(\theta) = p_{x_1,\dots,x_n}(x_1,\dots,x_n;\theta) = \prod_{i=1}^{n} (K+4\theta)\, x_i^{4\theta+\frac{1}{2}}$$

Define $L(\theta) = -\log(\ell(\theta))$

$$L(\theta) = -\sum_{i=1}^{n}\left[\log(K+4\theta) + \left(4\theta+\tfrac{1}{2}\right)\log(x_i)\right]$$

To find the maximum likelihood estimator $\hat{\theta}$ we minimize $L(\theta)$

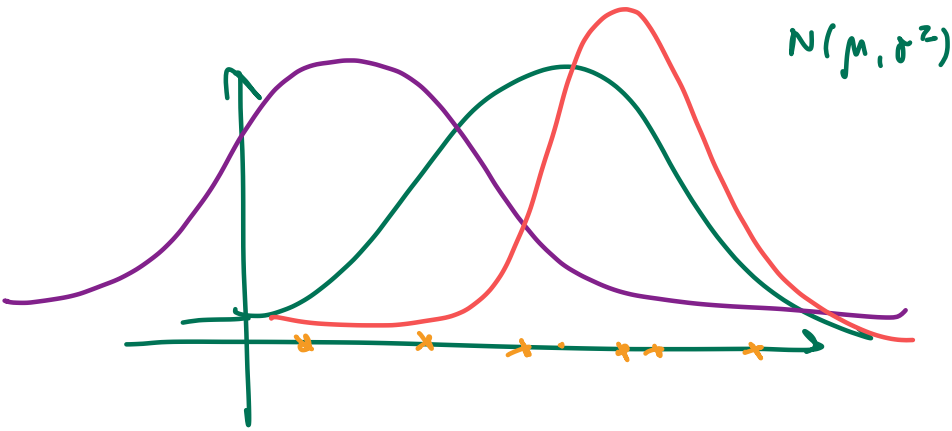
minimum condition

$$L'(\theta) = -n\frac{\cancel{4}}{K+4\theta} - \sum_{i=1}^{n} \cancel{4}\log(x_i) = 0$$

derivative

$$\Leftrightarrow \frac{-n}{\sum \log x_i} = K+4\theta \quad \Leftrightarrow \quad \theta = -\frac{1}{4}\left(K + \frac{n}{\sum \log x_i}\right)$$

$= \frac{3}{2}$ from a)


$N(\mu, \sigma^2)$

$L(\theta)$
λ, θ

EXERCISE 3

a) NOTE: here integration by parts is not super straightforward. If integration by parts is in an exam question, I will provide you with the formula.

$$\int_0^{+\infty} k\theta^3 x^2 e^{-\theta x}\, dx = 1$$

$$\int x^2 e^{-\theta x}\, dx \overset{\text{BY PARTS}}{=} x^2 e^{-\theta x}\left(-\frac{1}{\theta}\right) - \int 2x e^{-\theta x}\left(-\frac{1}{\theta}\right)$$

$$= -\theta x^2 e^{-\theta x} + \underbrace{\int 2x e^{-\theta x} \frac{1}{\theta}\, dx}_{\text{by parts}}$$

$$2x\, e^{-\theta x}\left(-\frac{1}{\theta^2}\right) + \underbrace{\int 2 e^{-\theta x}\left(+\frac{1}{\theta^2}\right) dx}_{= -2\frac{1}{\theta^3} e^{-\theta x}}$$

Hence:

$$\int_0^{+\infty} k\theta^3 x^2 e^{-\theta x}\, dx = k\theta^3 \left[x^2 e^{-\theta x}\left(-\frac{1}{\theta}\right) + 2x e^{-\theta x}\left(-\frac{1}{\theta^2}\right) - 2\frac{1}{\theta^3} e^{-\theta x} \right]_0^{+\infty}$$

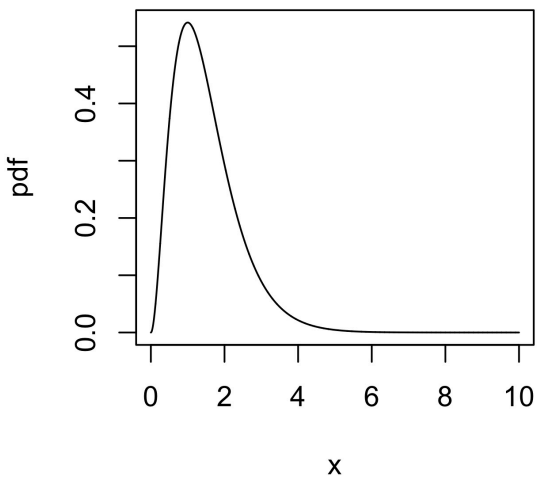
$$= k\theta^3 \left[0 + 0 - 0 - 0 - 0 + \frac{2}{\theta^3} \right] = 1$$

$$\Rightarrow k = \frac{1}{2}$$

b)

```
theta = 2
k = 1/2
x = seq(0,10,0.001)
pdf = k*theta^3*x^2*exp(-theta*x)
plot(x,pdf,type="l")
```

try some values of θ



my sketch done by hand

c) Similarly to <u>exercise 1</u>

$$l(\theta) = \prod_{i=1}^{n} K\theta^3 x_i^2 e^{-\theta x_i}$$

$$L(\theta) = -\sum_{i=1}^{n} \left[\log K + 3\log\theta + 2\log x_i - \theta x_i \right]$$

We set $L'(\theta) = 0$ to find the minimum.

$$L'(\theta) = -\frac{3n}{\theta} + \sum x_i = 0 \iff \theta = \frac{3n}{\sum_i x_i}$$